CE211 COMPUTER HW & MAINTENANCE I

CHAPTER 3
MOTHERBOARDS

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Sub-topics

3.1 Introduction
3.2 Form factors of motherboards
3.3 CPU and chipsets
3.4 Packaging of RAM modules
3.5 Types of expansion cards
3.6 On-board ports

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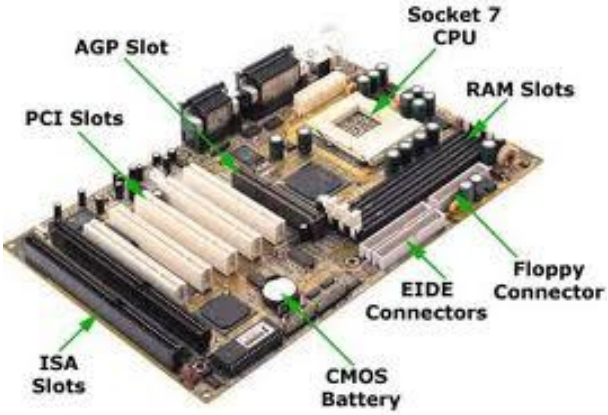
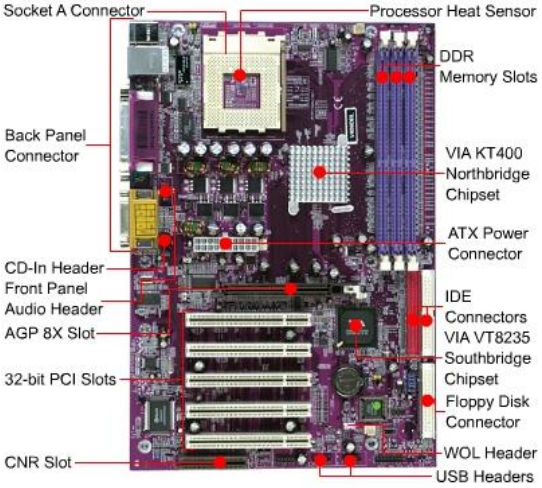
3.1 Introduction





Also known as system board, the motherboard is the main PCB (Printed Circuit Board) that holds the majority of PC's system components. It is said to be the nervous system of the PC, making it possible for the CPU to communicate with all system components (devices).

The common components found on the motherboard are:

- CPU sockets or slots
- RAM slots memory modules e.g. DIMMs, SIMMs.
- Expansion slots for expansion slots
- Power cables, power connectors and ribbon (data) cables
- Ports for I/O devices or peripherals e.g. mouse, keyboard, printer
- Microchips (ICs) e.g. ROM BIOS, CMOS RAM, controller chips, chipset
- RTC (Real Time Clock)
- Lithium battery for CMOS RAM
- Jumpers for configuration e.g. of motherboard and password.
- LEDSs for e.g. internal speaker, reset button, power button.

Note:
Most motherboards contain some built-in components (e.g. video, sound, network adapter) with built-in ports (e.g. PS/2, USB).

Design issues of motherboards

1. Expandability

Parts whose specification changes much tend to remain in sockets or slots e.g. CPU, RAM, video card, sound card and SCSCI controller.

2. Low cost
Parts not needed much by user are left out of base specification to keep the cost down e.g. NIC, modem and TV card.
3. Upgradeability
Future upgrading is considered by leaving empty slots for new technologies e.g. PCI cards, AGP cards, RAM modules.
4. Standardization
Made to given specification so that it can be serviced or repaired i.e. open systems or open specification.
5. Cooling system
There should be proper air-flow in system to cool components.
CPU needs cooling system i.e. heat sink & electric fan.
6. Space efficiency
Utilize as much space as possible e.g. expansion slots arranged in a given order.

3.2 Form factors of motherboards

Form factors refer to the different dimensions (sizes) and styles (layouts) of the motherboard available (or ever existed). They determine the type of system unit (case) the motherboard can fit into.
The 2 prominent form factors of motherboards are:

1. AT (Advanced Technology) are obsolete (no longer in use).
2. ATX (Advanced Technology Extended) are still in use.

Variants (versions) are:

1. AT has Full AT, Baby AT (mini AT & micro AT) and LPX
2. ATX has Full ATX, mini ATX, micro ATX, NLX and WTX

Note:

- IBM's PC/XT model (first desktop IBM PC) was the first form factor of motherboards, but became obsolete when AT form factor was introduced.
- IBM used open specification which made PC/XT form factor a standard motherboard at that time.

Comparison of AT and ATX

Components	**AT**	**ATX**
Keyboard	5 pin large DIN connector	6 pin mini connector PS/2
Power connector	Single row, 2 connectors	Double row, single connector
Expansion slots	Mainly ISA slots	ISA, PCI and AGP slots
External connectors	Minimal keyboard and mouse	Variety of ports serial, parallel, USB

Summary of sizes

Form factor	inches	millimetres
Full AT	13.8 x 12.0	351 x 305
Baby AT	10.0 x 9.0	254 x 229
Full ATX	12.0 x 9.6	305 x 244
Mini ATX	11.2 x 8.2	289 x 208
Micro ATX	9.6 x 9.6	244 x 244
Flex ATX	9.0 x 7.5	229 x 191

Proprietary motherboards
Proprietary means not standard (not an industry form factor).
Used in low-profile, slim case system; but can also be found in tower cases.
Companies which use proprietary motherboards include Compaq and Packard bell.

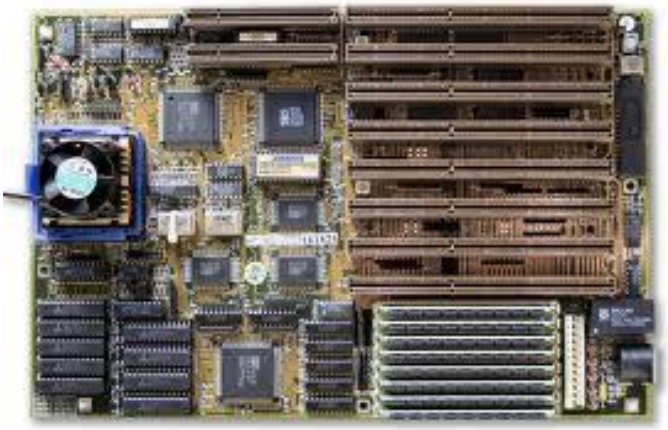
Problem
It is not usually easy to upgrade and also very expensive to repair. Why?
Motherboard case does not fit and power supply usually different.
<See LPX motherboard ahead>

Full AT
Big size (13.8 x 12.0 inches or 351 x 305 mm) motherboard with only 2 ports (for keyboard and cassette as storage device).

IBM AT motherboard

Problems
1. Upgrading it is impossible because of its large size (now obsolete).
2. Lack of external connectors (ports) for other devices.

Baby AT
Smaller size than Full AT became popular until 1996.
Essentially similar to Full AT, except size (maximum 10.0 x 9.0 inches or 254 x 229 mm).
Processor socket was located at the front of the motherboard.
Full length expansion cards extended over the processor.
Had 2 versions e.g. mini AT, micro AT.

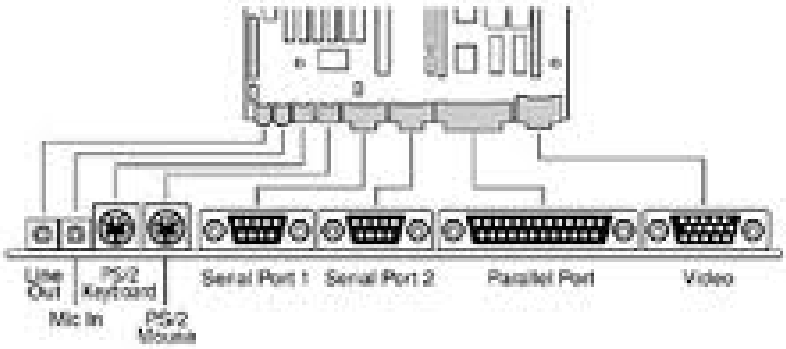


Features
1. Internal card plug directly into motherboard.
2. Externally, only one on-board port (for keyboard) visible.

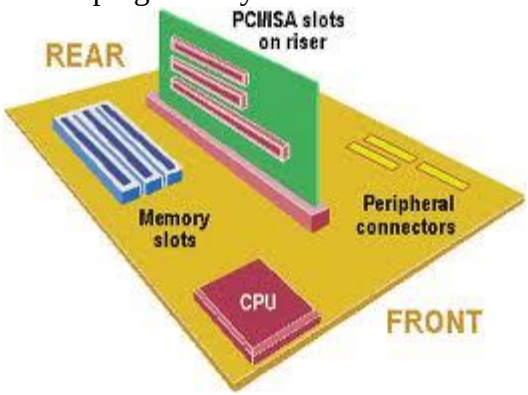
Problems
1. Not easy to access processor because you need to first remove some or all expansion cards
2. Power supply only 5V and 12V, but some components need 3.3V eg CPU and PCI cards. It means additional built-in regulators added to motherboard.
3. Heat dissipation a problem, since Pentium speeds had increased. It means needed a fan and a heat sink.
4. Limited in expansion slots and driver bays.

LPX
LP stands for Low Profile i.e. used in low profile desktop systems
(Slim cases). Designed (1987) by a company called Western Digital, now manufacturers of HDDS). Specialized variant of Baby AT (overall smaller sizes). This has also been duplicated by many motherboard manufacturers.

External features
Standard single connector panel at rear (back) of MOTHERBOARD:
1 video (VGA DB15), 1 Parallel (DB25), 2 serial (DB9) and mini din (PS/2) for keyboard and mouse.



Internal features
Central riser card that plugs into motherboard. The riser card has expansion slots (on one or both sides) which are parallel to the motherboard.
Expansion cards plug horizontally (sideways) into central riser card. They do not plug directly into the motherboard.



Some other motherboard designs also use riser cards, but have double connector panel at the bottom of rear (back).

Problems
1. Semi proprietary (not standardized) ie not interchangeable between manufacturers.

2. Cannot fit big size of Pentium II/III processor, because of slim case.
3. Cannot provide adequate cooling for Pentium II/III processors, because riser card prevents airflow within the system. Additional fans are needed.
4. Difficult to remove the motherboard for replacement, because of riser card.
5. Cannot fit newer bus e.g. AGP

ATX

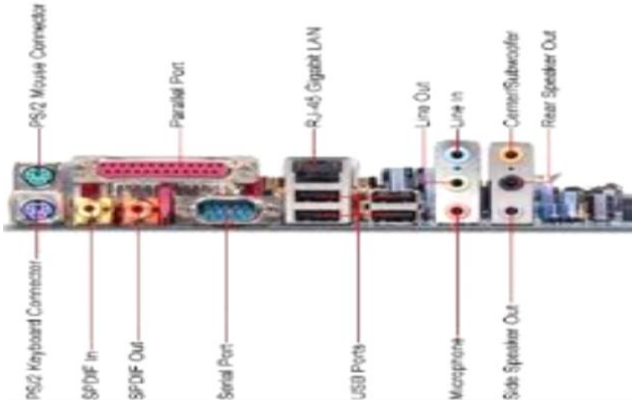
Designed by Intel (July 1995) with new enhancements added, combined best features of Baby AT and LPX.
As small as Baby AT and incorporates all external connectors systems have today.
Intel used open specification for other manufacturers. Most popular today and recommended.
Has versions i.e. full ATX, mini ATX, micro ATX, flex ATX

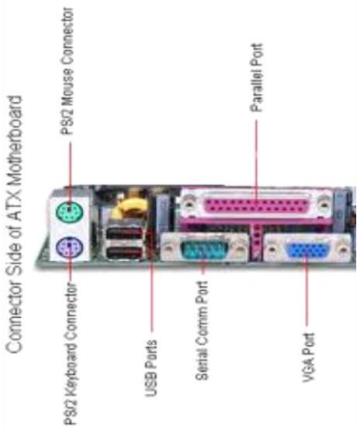
Improvements

1. Modified power supply location and added 3.3V for some components (On addition to 5V and 12V on MB). No built-in regulator on MB.
2. CPU sockets and voltage regulator moved to the right hand side of the expansion slots.
3. Reduced size and integrated components e.g. flash BIOS, I/O logic chips and keyboard controllers. This has created more space for CPU.
4. Relocated CPU and RAM to improve on cooling (better airflow), because power supply fan is used and heat sink has enough space.
5. There is also easy access to CPU and RAM so no need to interfere with expansion cards during installation.
6. Relocated internal I/O connector e.g. floppy disk and hard disk are near drive bays.
7. This made shorter internal cables and no need to remove expansion card or drive to access connectors. This also led to lower cost of manufacturing.
8. Double high external I/O connector panel meant no cables running from internal connectors to the rear (back).

External features

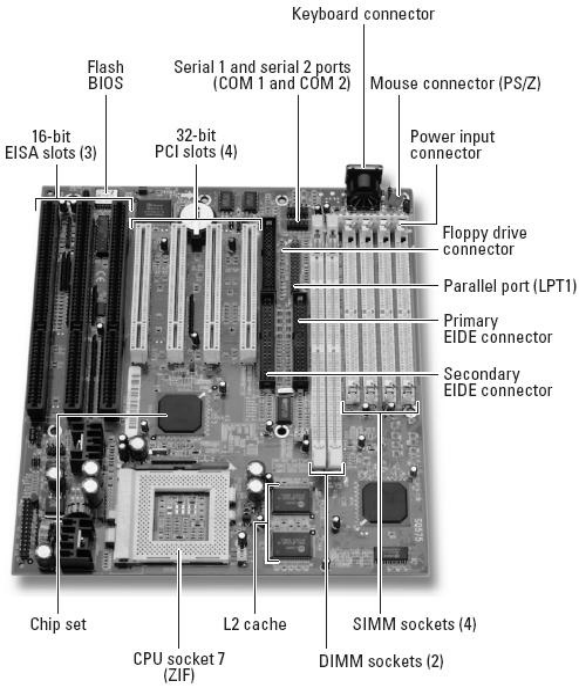
Unique double high connector panel for built in connectors on motherboard.





Internal features

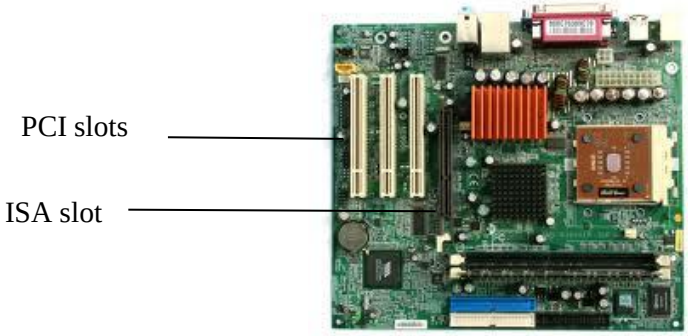
Expansion cards plug directly into the motherboard ie no riser card like in LPX
A total of seven I/O expansion slots (eg 4 PCI & 3 EISA as seen below) on the motherboard.
Power supply connector is double row and single connector.



Mini ATX
Simply a smaller version of full ATX; no major differences. Both ATX and mini ATX cannot be used in AT design cases.

Micro ATX
Designed by Intel (December 1997) for smaller, low cost systems with limited expansion capability. Smaller than mini ATX. Can fit into standard ATX case or one of the new micro-tower desk top designs i.e. backward compatibility with ATX

Improvements aimed at low costs
-Reduced size compared to ATX or mini ATX
-Reduced number of I/O expansion slots from 7 to 4 (e.g. 3 PCI & 1 ISA as seen below).
-Uses a smaller power supply e.g. SFX design.

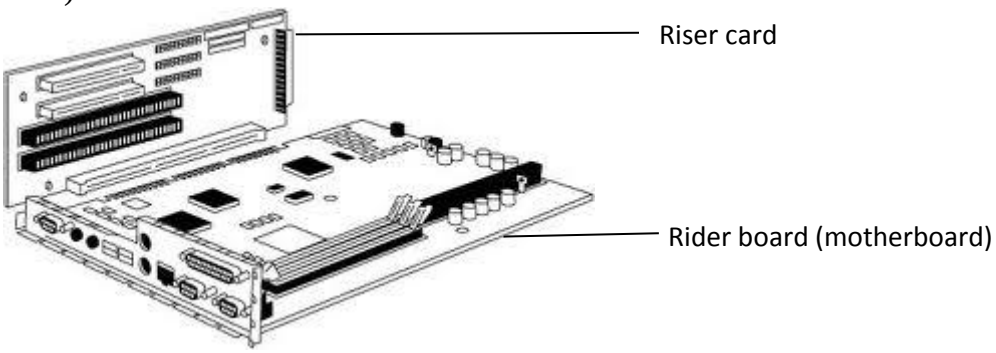


Flex ATX
Designed by Intel (March 1999) to replace micro ATX.
Intended to allow a variety of new PCs, smaller than micro ATX.
Only addressed requirements of the motherboard, but not the overall system eg details of interface, memory or graphics were left out to the system designers.
Only supports sockets for processors eg socket 7 and socket 370.
Allow backward compatibility with both ATX and micro ATX.
Improvements
1. Lower overall production costs.

2. Facilitates design of smaller systems.

NLX
NLX stands for New Low Profile, replaced proprietary LPX.
Designed (1996) by Intel for maximum flexibility and space efficiency.
Has numerous improvements to allow full integration of latest technologies.
Improvements
1. Fully standardised (unlike LPX), making it easy to service and repair.
2. Supports all desktop processors eg can fit big size of Pentium II/III, (including SECC technologies) and can adequately cool them.
3. Supports new technologies (AGP, USB) and new memory modules (DMM and RIMM).

New features
1. Rider board (motherboard) plugs into riser card (not riser card into motherboard as for LPX).



2. Has no internal cables (eg drive cables) and connectors (power supply cables) attached to it.

Advantages
1. Because rider board plugs into riser card
   Easier to remove motherboard without affecting the riser card or expansion cards.
2. Because of no internal cables and connectors
   Very easy to remove motherboard without unplugging cables and connectors.
   Easy to remove power supply or disk drive without removing boards.

WLX
WLX stands for New Low Profile; which replaced proprietary LPX.
Designed by Intel (1998) for maximum flexibility and space efficiency.
Has numerous improvements to allow full integration of latest technologies.
Designed to support:
- 32-bit and 64-bit Intel processors
- Dual processor motherboard
- Future memory technology
- Future graphics technology
- Tower form factor

Special features
1. Flex slot I/O cards (double wide PC).
2. Pull out drawers and swinging side panels.
3. Two power supply form factors (single fan 350W & dual fan 850W).

Advantages
1. Easy access to internal components eg memory slots and expansion slots.
2. Allows larger, more power hungry and malfunction cards.
3. Reduced E/M interference, since I/O signals far away from processor, chipset and memory.

3.3 CPU and chipsets
CPU
Also referred to as microprocessor or simply processor.
Refer to Chapter 1 for:
- PGA and ZIF for sockets
- SECC and SEPP for slots
- What to consider in designing a CPU.

Intel Core 2 Duo Processor has 775 pins (1/2 provide only power).

How to install ZIF CPU
CPU installation involves:
- Lever on the CPU-socket open when upright.
- Heat sink and cooling fan on top of the CPU.
- Metal clips that hold heat sink tightly on the CPU.

1. Align 2 marked similar corners (without pins) of CPU and insert it carefully into the socket on the MB.
2. Use ZIF (Zero Insertion Force) to let the CPU go into the socket ie without applying much force.
3. Gently lower the lever to hold the CPU in place.
4. Apply a thin coating of heat sink compound directly to the top of CPU.
5. Place the heat sink and electric fan on top o0f the CPU.
6. Attach metal clips to hold heat sink and fan tightly down against CPU.

<See diagrams below for illustrations>

Cache
This is small memory that provides high speed access by CPU.
There 2 main types of cache:
1. Primary cache (Level 1 cache)
2. Secondary cache (Level 2 & 3 cache)

Primary cache (Level 1 cache)
This is located inside the CPU (internal cache) for faster and easy access of data by the CPU.
It is used to store recently accessed data (e.g. web pages) and commands.

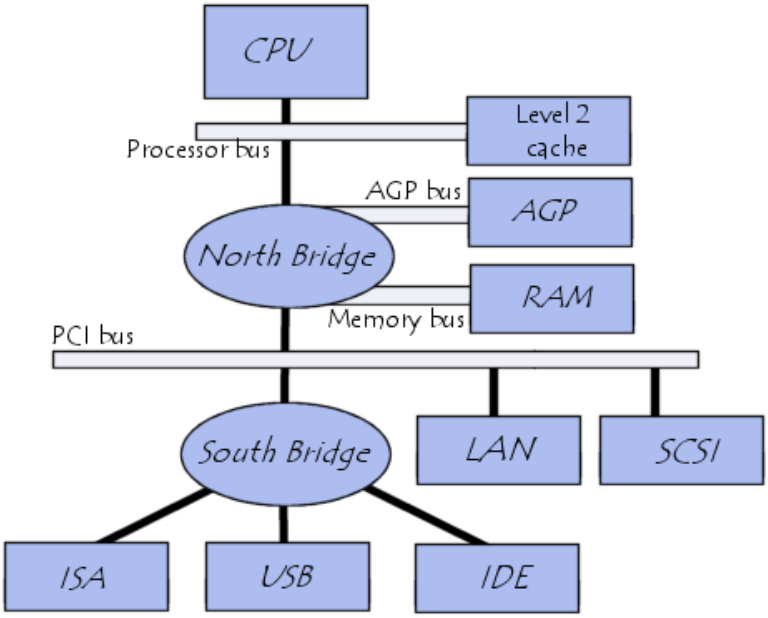
Secondary cache (Level 2 & 3 cache)
This is located outside the CPU (external cache) and more than level 1.
It holds copies of programs and data from main memory.
Level 2 was initially placed on motherboard, but it now found within CPU.
While level 3 (used by modern PCs) is always found outside CPU i.e. between CPU and main memory.

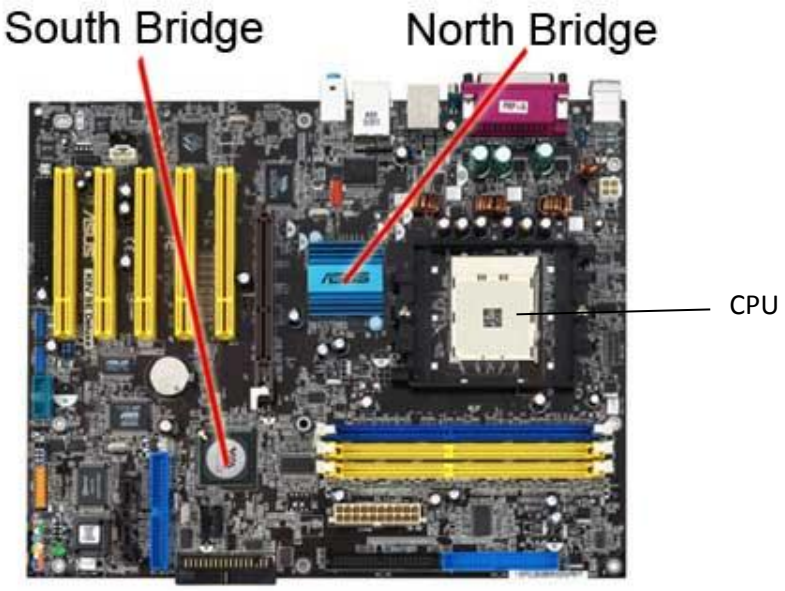
Chipsets
What is a chipset?
It is a single chip with many functions making a glue logic.
It is a set of specialized chips on the motherboard or expansion card.
It has a bus controller and system timer, among other things.
It is usually designed to work with a specific family of microprocessors.
It controls communications between the CPU and external devices; makes it to determine system performance.

Chipset manufacturers
Initially Chips & technologies, VLSI, Acer, Erso, Opti, Sunta and UMC.
Many of these fell out because of stiff competition.
Cos. in chipset manufacturing today are Intel (with 90% of chipset market), Ali (Acer Laboratories Inc.), VIA Technologies and SiS (Silcon Integrated Systems).
Intel (1994) entered the market of chipset manufacturing, because Companies were taking long to produce chipsets to support their new CPUs 286 and 386. Eventually Intel started producing all the 3 i.e. CPUs, chipsets and MBs. The manufacturer of a chipset is often not manufacturer of the motherboard.

Why a chipset was required in a PC
Before 1986, motherboards had many chips to handle different functions.
Later, more than 100 chips were needed to handle increased functions.
Chipset CS8220 (with 5 chips) was the 1st chipset made by Chips & Technologies.
Chipset CS8221 or NEAT (New Enhanced Advanced Technology) was the 1st chipset made for IBM PC AT (early PCs) by Chips and Technologies for the Intel CPU 80286.

Modern chipsets are based on Intel Pentium CPUs and are of main types:
1. North bridge
2. South bridge





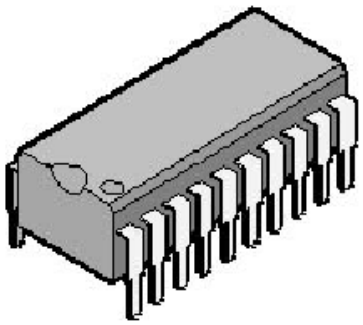
North Bridge
It is the main component of the motherboard, placed near CPU.
It runs at full speed of the motherboard i.e. connects CPU to very high-speed devices e.g. RAM (main memory) and AGP (graphics controllers).

South bridge
It is for lower speed components of CPU i.e. connects lower-speed devices to North Bridge through PCI bus e.g. ISA, USB, IDE (hard disk controllers), BIOS and CMOS.

3.4 RAM modules
RAM chips are packaged in different ways and they differ in the number of pins, notches and chips. These are:

1. Obsolete DPP & SIPP
   DIPP (Double Inline Pin Package)
   SIPP (Single Inline Pin Package)
2. Older SIMM
   SIMM (Single Inline Memory Module)
3. Newer DIMM & RIMM
   DIMM (Double Inline Memory Module)
   RIMM (Rambus Inline Memory Module)

DIPP (Dual Inline Pin Package)
The 1st RAM packaging used in old IBM PC AT. Obsolete (no longer in use today) Used by older PCs e.g. in IBM AT. Had 2 rows of pins for sockets in MB (8 chips per row). Arranged in rows of usually 8 chips per row, but sometimes 9 chips per row (9th chip for parity error checking). Packed in tubes usually each with 8 to 16 chips and marked with speed and size e.g. AB256-80 means 256 Kb (kilobit) and access time of 80 ns (nanosecond).



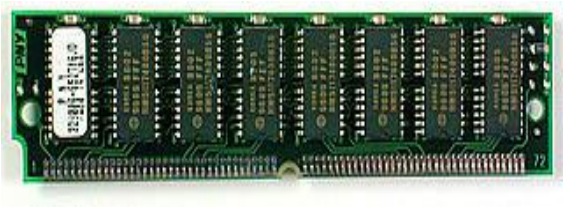
Example
8 chips @ 256Kb means total RAM = 8x256 Kb = 256 KB (kilobytes)

Problem
Processor faster than the memory and so memory gets overrun with requests from the processor, causing serious performance problems.

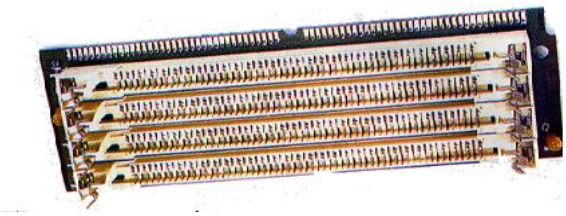
Solution
Manufacturers introduced wait states into RAM i.e. Processor waits one or more clock cycles, allowing the RAM to catch up. Wait state of zero means that the processor and the memory are equally matched in speed.

SIPP (Single Inline Pin Package)
Obsolete type, no longer in use today.
In form of sticks in banks, compared to multiple DIPPs.
Each bank had 2 sticks and each stick had DIPPs soldered together.

SIMM (Single Inline Memory Module)
Old type, no longer in use (unless on old systems).
Bus width = 16 or 32 bits, and no. of pins =30 or 72.
With no notch (30 pins) or one notch (72 pins).



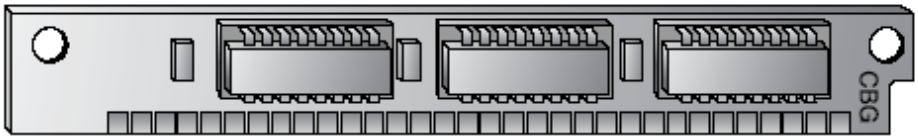
SIMM module (1 notch, 72 pins)



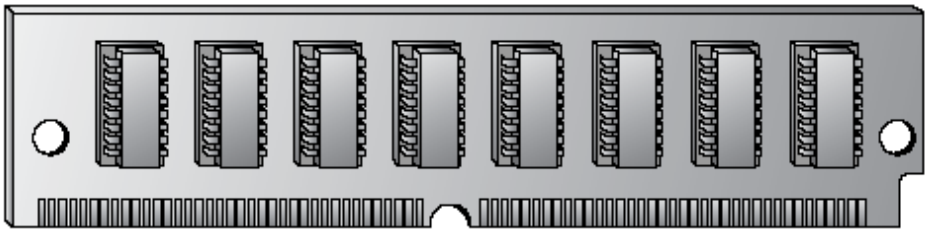
SIMM slots on motherboard

The next type of RAM packaging that is commonly seen in computers is called the Single Inline Memory Module (SIMM). SIMMs were developed because DIPs took up too much "real estate" on the logic board. Someone got the idea to put several of the DIP chips on a small circuit board and then make that board easily removable.

Single Inline Memory Modules (SIMMs)



30-pin SIMM



72-pin SIMM

The first SIMMs had nine small DIP chips on them and took up less room than before, because four of them could be installed in the same space as one row of the older DIP memory chips. In order to accomplish this, the SIMMs are installed very close to each other at approximately a 45º angle. This design was also meant to prevent "**chip creep**"—whereby the chips that have been placed in sockets on the board start to slowly move out of their sockets (caused by the repeated heating and cooling of the system board).
An old technician's trick: If an older computer (PC or XT) is having strange, irreproducible problems, open the case and reseat all socketed chips by pressing them down securely in their sockets. Most of the time that will solve the problem. If it does, then the problem was caused by "chip creep".
Most memory chips are 32-bit; so are several of the processors. You have a problem, however, when you have 32-bit memory chips and a 64-bit processor. To solve this, you must either install the SIMMs in pairs (always installing multiples of two and this is especially true for Pentium computers) or change to a **DIMM** installation.

DIMM (Dual Inline Memory Module)
Newer type, still in use in most PCs.
Double sided memory chips, twice the no. of chips
Bus width = 32 or 64 bits, and no. of pins =72 or 168.
With one notch (30 pins) or two notches (72 pins).

DIMM module (2 notch, 72 pins)

DIMM slots on motherboard

The final type of memory package is known as a **DIMM** (Dual Inline Memory Module). DIMMs are dual-sided memory chips that hold twice as many chips as a SIMM. (And, except for the fact that they have chips on both sides, they look just like a SIMM.) Generally, the DIMMs you'll run into will have either 72 or 168 pins. Some DIMMs are 32-bit, but more and more are 64-bit and only have to be installed one at a time in Pentium-class computers. DIMMS are a real improvement over older memory modules. They provide larger amounts of RAM on a single module and are easy to install. They slide straight down into a slot and are secured by a pair of locks that swing into place above the card as it seats fully in the slot. Check the motherboard manual or the vendor's Web site for the approved list of DIMM modules. There are many variations in electronic design (parity, non-parity, etc.), and you must make sure that the DIMM will actually work with the combination of motherboard and CPU you are working on. Just because the card fits does not mean it will work.

RIMM (Rambus Inline Memory Module)

Rambus is a trade name, not acronym (abbreviation). RIMM has a 184-pin connector and other versions do exist e.g. Samsung RIMM. It is the newest RAM module used in new PCs. It was developed by Kingston Technology Corp.

Direct Rambus is a relatively new and extremely fast (up to 800MHz) technology that uses, for the most part, new methodologies in memory system design. Direct Rambus is a memory bus that transfers data at 800MHz over a 16-bit memory bus. Direct Rambus memory models (often called RIMMs), like DDR SDRAM, can transfer data on both the rising and falling edges of a clock cycle.

That feature, combined with the 16-bit bus for efficient transfer of data, results in the ultra-high memory transfer rate (800MHz) and the high bandwidth of up to 1.6GB/second (more than twice that of 100MHz SDRAM).

RIMM is not an acronym and is a trademark of Rambus incorporated for the Direct Rambus or DRAM modules. RIMM is a type of computer memory that resembles DIMMs; however, it is 184-pin and is available with built in ECC support and Non-ECC at speeds up to 800MHz. RIMM modules are commonly used on the Intel Pentium 4 motherboards.

Unlike most other types of computer memory, computers that support RIMM require a continuous signal. If a memory socket is left empty, the computer will not work properly. Therefore, users must utilize C-RIMM modules in any slots that do not have RIMM modules. The memory chips themselves come in many different types of packages. The ones most frequently encountered are discussed in the following sections.

How much RAM capacity is required for the PC?
Depends on software (OS or applications) to be installed on the PC.
For example, Microsoft recommends a minimum of 128MB for Windows XP, but it is better to have 256MB or even 512MB. Older versions of Windows required far less memory e.g. Windows 98 (64MB) and Windows 2000 (128MB). What about Windows 7 & 8 now? Modern PCs have 2 to 4GB of RAM. Why? See adverts.

What to consider when upgrading RAM

1. System hangs often and may never open application. RAM is little.
2. Find out the type of RAM in PC and upgrade with this same type.
3. Consider the speed of the previous module and the band width.

How to install DIMM memory module

1. Locate a free RAM slot and align the RAM module with the slot.
2. Align RAM module (use notches) in RAM socket (use alignment key).
3. Insert the RAM module into the slot and make sure it clicks into place.
4. Lock the RAM module into place using clips into side notches.
5. Close the systems case and power the computer
6. Check in BIOS/CMOS setup to confirm new RAM is recognised.

Precautions!

1. Switch off the computer to avoid electrocution.
2. Discharge yourself off ESD to avoid spoiling chips.

See Types of RAM in another chapter e.g. DDR, DDR2 & DDR3 RAM.

3.5 Expansion cards
These are smaller PCBs that fit into expansion slots on motherboards.
They add functions to the PC giving it added capabilities. They include:

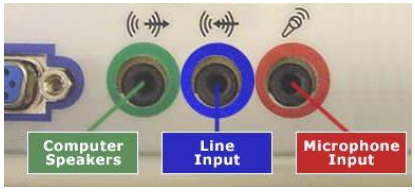
1. Sound cards
2. Video cards
3. Internal modem
4. NIC (Network Interface Card)
5. TV tuner cards

Sound cards
Provide output (e.g. speakers, earphones) and input (eg microphone) of audio (sound).
Many motherboards have on-board sound card (built in the motherboard). But is better to have to buy and install a modern sound card.

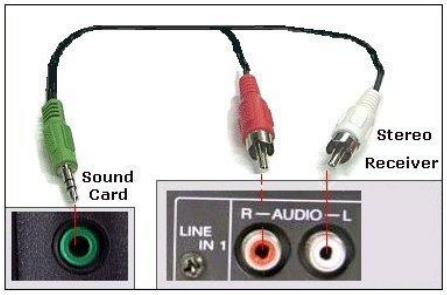
For example, for music production use modern sound card for better sound quality and other features.
A sound card translates signals into sounds that can be played back through speakers.
A PC sound card is installed into the PCI slots on motherboard.

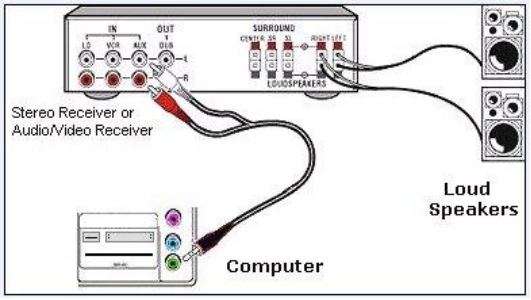
Most sound cards have 3 BNC (round) colour coded jacks (ports) as:
1. Line out (**green** with arrow pointing outward) for speakers
2. Line in (**blue** with arrow pointing inward) for CD player and cassette tape).
3. MIC in (**pink** with picture of microphone) for microphone.

Note: Another port is DB15 female (game port).

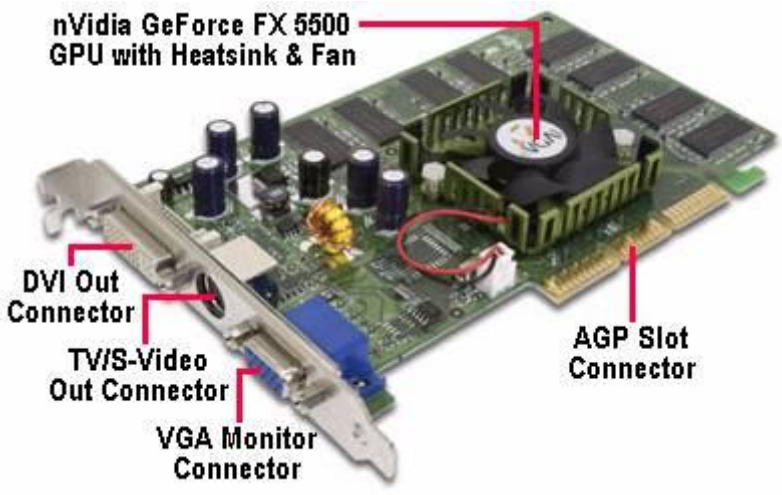


Connecting PC to a stereo receiver (radio)
1. Plug single end of Y adapter cable into Line out of sound card at back of PC.
2. Plug in the other two ends of Y cable into jacks of stereo receiver.
   Match red plug of Y cable to red jack on stereo receiver and connect.
   Match black (or white) plug of Y cable to white jack on stereo receiver.

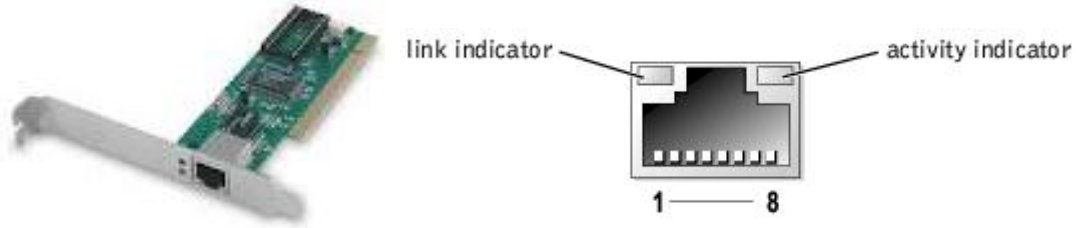




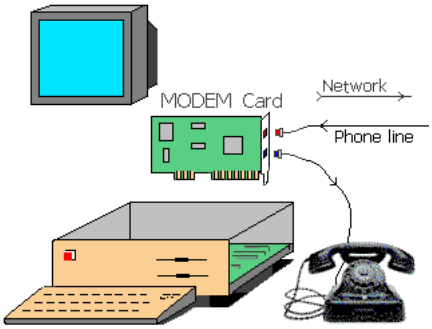
Video cards Provide visual (graphics) output to screen of a monitor or other video device.
Some video cards also allow input from a video camera, VCR, DVD player, or other video sources.



NIC (Network Interface Card)
Allows a PC to be connected to LAN (Local Area Network).

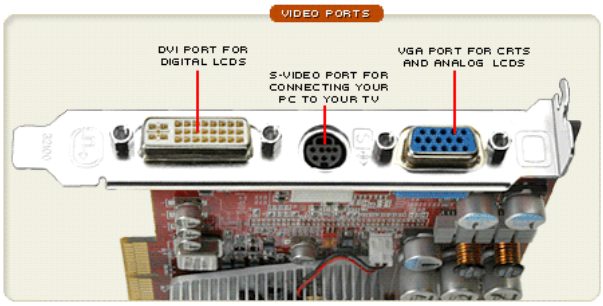


Internal modem
Allows a PC to be connected to Internet.

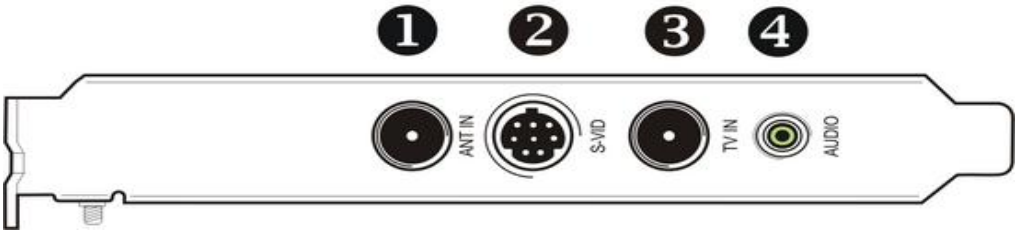


TV tuner cards
TV tuner cards allow TV signals to be received by a PC (television tuner). Most TV tuners also function as video capture cards, allowing them to record TV programs onto a hard disk.

A TV tuner card is installed into the PCI slots or the newer PCI Express (PCIe) on motherboard.
Some TV cards contain a tuner and an analog-to-digital converter along with demodulation and interface logic.



Common TV tuner ports



1. Antenna (coaxial connection)
2. Video connection
3. Cable/Antenna (coaxial connection)
4. Stereo audio mini jack connection

Installation of PCI card

1. Align PCI card (use notches) in expansion slot (use alignment key).
2. Press it firmly into the slot so that it sits properly.
3. Check in BIOS/CMOS setup to confirm installation is properly done.

3.7 On-board ports

Some components are built in components were available on expansion cards on earlier PCs. The most common of are video card, sound card, NIC and modems. On-board ports are therefore those found on the motherboard e.g. PS/2 ports for keyboard and mouse, BNC ports for sound card, DB9 port for video card, DB25 port for printer and USB ports for many devices. Just look at the back of the PC to determine the functions that are built-in.